

Detecting Multiple Independent Events

Independent Events

- In the special case where $P(E|F) = P(E)$, E is independent of F . Knowledge that F has occurred does not change the probability that E occurs

$$P(E|F) = \frac{P(EF)}{P(F)} = P(E) \iff P(EF) = P(E)P(F)$$

- Since the equation is symmetric, if E is independent of F , so is F independent of E
- Two events are said to be independent if

$$P(EF) = P(E)P(F)$$

Two events that are not independent are **dependent**

Example

A card is selected at random from an ordinary deck of 52 playing cards. If E is the event that the selected card is '10' and F is the event that it is a 'heart', show that the two events are independent.

$$P(E) = \frac{4}{52}$$

$$P(F) = \frac{13}{52}$$

$$P(EF) = \frac{1}{52}$$

$$P(EF) = P(E)P(F)$$

Example

Two coins are flipped, and all 4 outcomes are assumed to be equally likely. If E is the event that the first coin lands heads and F is the event that the second lands heads, show that E and F are independent.

$$P(E) = P\{HT, HH\} = \frac{1}{2}$$

$$P(F) = P\{TH, HH\} = \frac{1}{2}$$

$$P(EF) = P\{HH\} = \frac{1}{4} = P(E)P(F)$$

Proposition: If E and F are independent, then so are E and F^c

Proof:

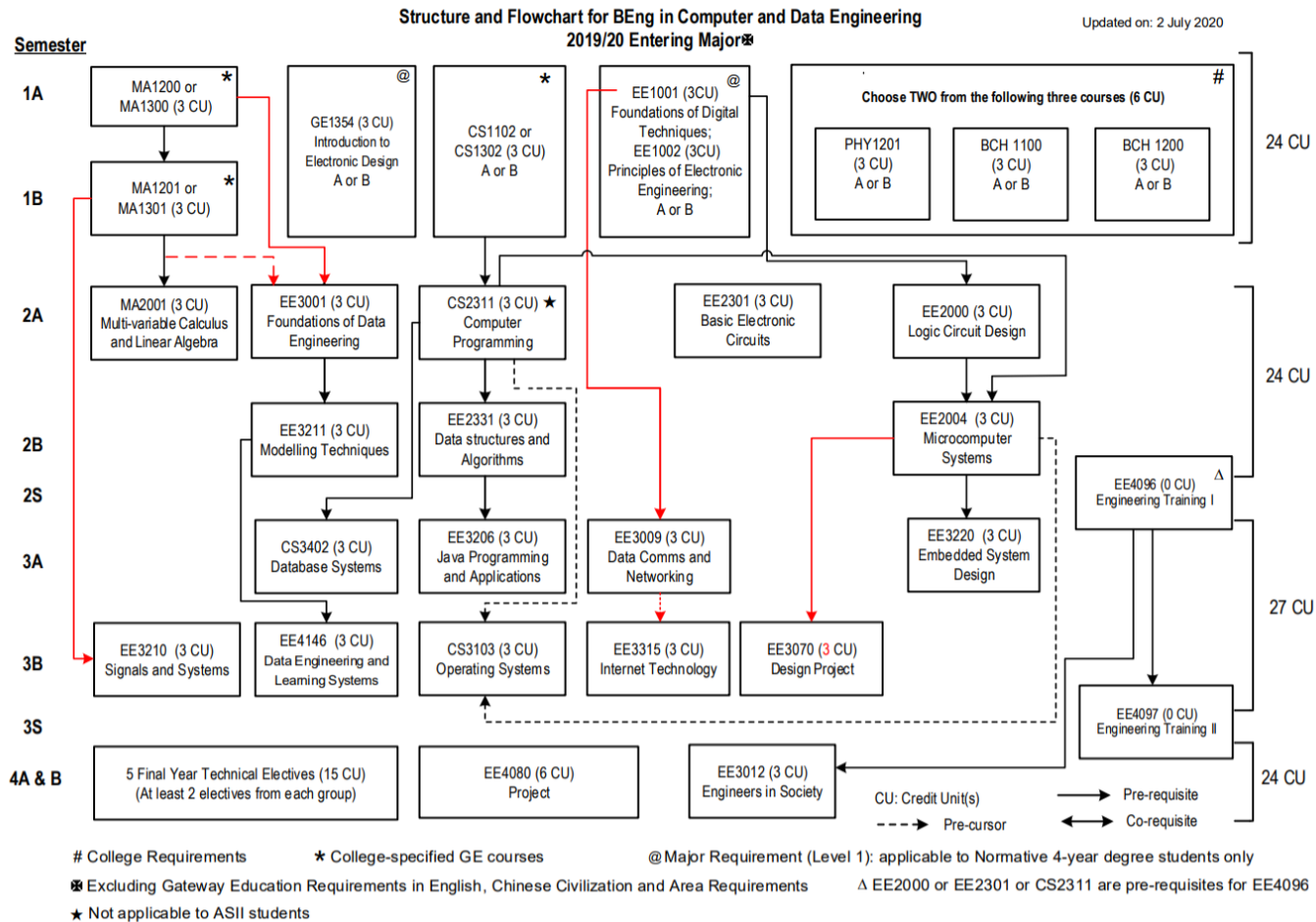
Assume that E and F are independent. Since $E = EF \cup EF^c$, and EF and EF^c are obviously mutually exclusive,

$$P(E) = P(EF) + P(EF^c) = P(E)P(F) + P(EF^c)$$

or equivalently,

$$P(EF^c) = P(E)(1 - P(F)) = P(E)P(F^c)$$

Example



Is the grade of EE2301 and EE3315 independent?

Suppose E is independent of F and is also independent of G . Is E necessarily independent of FG ? The answer, somewhat surprisingly, is no.

Example

Two fair dice are thrown. Let

E event that the sum is 7

F event that the first throw is 2

G event that the second throw is 5

Show that E is independent of F , and E is also independent of G , but E is not independent of FG .

E and F are independent because

$$P(E) = P\{(1,6), \dots, (6,1)\} = \frac{6}{36} = \frac{1}{6}$$

$$P(F) = P\{(2,1), \dots, (2,6)\} = \frac{1}{6}$$

$$P(EF) = P\{(2,5)\} = \frac{1}{36} = P(E)P(F)$$

E and G are independent because

$$P(G) = P\{(1,5), \dots, (6,5)\} = \frac{1}{6}$$

$$P(EG) = P\{(2,5)\} = \frac{1}{36} = P(E)P(G)$$

However, $P(E|FG) = 1 \neq P(E)$

Independence involving more than two events

Definition:

Three events E , F , and G are said to be independent if

$$P(EFG) = P(E)P(F)P(G)$$

$$P(EF) = P(E)P(F)$$

$$P(EG) = P(E)P(G)$$

$$P(FG) = P(F)P(G)$$

In general, the events $E_1, \dots, E_n$ are independent if for every subset $E_{1'}, \dots, E_{r'}$, $r \leq n$, of these events

$$P(E_{1'} \cdots E_{r'}) = P(E_{1'}) \cdots P(E_{r'})$$